

ABRAR FAHYAZ

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Education

University of Dhaka

B.Sc (Hons.) in Computer Science & Engineering

2022 - 2026

CGPA: 3.83/4.00 (Rank: 4/68)

Notre Dame College

Higher Secondary Certificate (HSC)

2019 - 2021

GPA: 5.00/5.00 (Top 10%)

Adamjee Cantonment Public School

Secondary School Certificate (SSC)

2017-2019

GPA: 5.00/5.00 (Top 3%)

Interests

- Biologically inspired systems; architectures that adapt, reorganize, or self-correct in ways reminiscent of organic cognition.
- Computational language: NLP, semantics, automata, and the foundations of language modeling.
- Distributed systems, decentralized/P2P communication models, and scalable computational infrastructure.
- Reinforcement learning, Vision Transformers, neural network optimization, and training dynamics.
- Cybersecurity, cryptography, digital forensics, binary and web exploitation, and adversarial reasoning through CTFs.
- Building complex systems from first principles—simulators, networks, editors—to understand underlying mechanics rather than abstractions.
- Modular software and web-based systems engineered through layered, incremental design.

Research Experience

Neuroplastic Neural Networks: Structural Plasticity and Hebbian Updates

2026

Springer Nature Computer Science

[Show Publication](#)

- First Author of an extended study of the conference paper under the supervision of Dr. Kaveh Kiani and Dr. Maybin Muyeba (*UofSalford*), who were judges at the ICDSAIA 2025.
- Includes an extended theoretical foundation, better statistical validation, a more detailed literature review, and a CNN expansion of the framework.
- Underwent two rounds of peer-review, and has been selected for publication as of March 2026.

Neural Networks with Neuroplasticity: Connection Pruning and Hebbian Updates

2026

ICDSAIA 2025, Part II, CCIS Vol. 2682, Ch. 5, pp 60-74

[Show Publication](#)

- Preliminary research into neuromorphic neural network architectures with adaptive learning capabilities.
- Integrated Hebbian updates, dynamic connection pruning and re-qualification and adaptive lr into a standard Neural Network for improved model efficiency.
- Conference paper presented at the International Conference on Data Science, AI and Applications (ICDSAIA 2025).

Investigating the Potential of NLP in Improving Sentiment Analysis Accuracy

2023

Presented at 1st DURS Undergraduate Research Summit 2023

- Conducted early experiments with Markov models and n-gram based statistical generation to probe how simple systems replicate syntactic and tonal patterns in human language.
- Developed a lightweight sentiment analysis pipeline for the IMdB dataset, integrating a custom TF-IDF vectorizer and multiple preprocessing layers to capture linguistic nuance.
- Selected as one of the top 26 research papers at the DURS summit; later packaged the model into a functional software tool (`N_avram`) and presented it at TMT-P, earning an award.

Projects

Neural Connectome of *C. elegans*

[Source Code](#)

- Implemented a Leaky Integrate-and-Fire (LIF) Spiking Neural Network using NumPy and OpenWorm datasets.
- Modeled all 302 neurons and 5100+ synapses (chemical + gap junctions) of *C. elegans*.
- Mapped sensory, inter, and motor neurons to input-hidden-output layers to emulate full connectome dynamics.

ChaOS: Turing-Complete OS Emulator

[Demo Video](#) — [Source Code](#)

- Built an OS emulator with virtual memory, process scheduling, I/O, filesystem, and shell.
- Designed a custom ISA and compiler toolchain enabling Turing-complete batch execution.

PlinyTheElder: Multi-Agent AI Tutoring System

[Source Code](#)

- Developed a multi-agent tutoring system with domain-specialized agents and tool integration.
- Implemented document ingestion for personalized tutoring via uploaded study materials.

N_Avram: Sentiment Analysis Framework

[Source Code](#)

- Built a sentiment classifier using Random Forest with TF-IDF, n-grams, and lemmatization.
- Achieved ~84% accuracy on IMDB, comparable to lightweight transformer baselines.
- Presented at Undergraduate Research Summit 2023 and 3MT (4th place).

Gravitational Physics Engine (N-body Simulation)

[Demo Video](#)

- Developed a modular N-body simulator using Newtonian mechanics with collision handling.
- Implemented configurable parameters (G, timestep, elasticity) and real-time visualization.
- Built Python and ThreeJS visualizers with interactive controls and performance optimization.

Strategic War Simulator

[Source Code](#)

- Built a military simulation platform with troop movement visualization and battle timelines.
- Integrated 100+ historical units and scenario persistence (JSON-based).
- Featured in [The Daily Star](#) for educational applications.

EasyForm AI [CSE3112]

[Source Code](#)

- Developed an LLM-powered document processing system for automated form filling.
- Implemented information extraction, form generation, and chatbot assistance.
- Achieved 90%+ accuracy and ~10x speedup over manual workflows.

P2P Video Streaming Platform [CSE3111]

[Source Code](#)

- Built a P2P streaming system using DHT-based discovery and TCP chunk exchange.
- Implemented progressive playback via 1MB chunking with integrity checks and redundancy.
- Developed FastAPI backend and Streamlit frontend for playback and file management.

Theia: Photo Sharing Platform [CSE2216]

[Demo Video](#)

- Developed a Flutter-based photo sharing app with Firebase backend.
- Implemented real-time sync, media capture, and multi-layer editing features.
- Added facial recognition, geo-clustering, and optimized large-scale gallery rendering.

Exoplanet Database [CSE2211]

[Source Code](#)

- Designed a normalized PostgreSQL schema for astronomical datasets.
- Built a Python backend and multi-parameter query interface with visualization support.

4-bit Adder/Subtractor [EEE2113]

Simulation

- Implemented a 4-bit adder/subtractor using basic logic gates.
- Supported signed arithmetic via 2's complement handling.

ShareTune Music Player [CSE2112]

Source Code

- Built a JavaFX-based music player with streaming over TCP.
- Designed a custom GUI with hand-crafted assets and animations.

ShN Text Editor [CSE1211]

Demo Video

- Developed a cross-platform C++/Qt text editor with syntax highlighting and compilation support.
- Implemented document analysis tools and integrated file explorer.

Professional Experience

Scale AI, Inc

September 2024 – Present

Advanced AI Coder

- Finetuned and refined technical code in Swift, Python, JavaScript, Java, C, and C++ for training and evaluating large language models.
- Developed and corrected complex code samples to improve model reasoning, tool usage, and multi-step problem-solving across multiple programming languages.
- Trained models and agents to recognize tools, reason about when and how to use them, and execute multi-agent workflows reliably.
- Designed adversarial prompts and stump tasks to identify model failure points, edge-case vulnerabilities, and systematic reasoning errors.
- Designed failure-driven training sequences to improve the model's understanding of ambiguous, complex, or multi-layered prompts.
- Contributed to OpenAI's Feather program shortly before the release of GPT-5, helping strengthen coding performance, reasoning depth, and tool interaction capabilities.
- Contributed as a Reviewer and QA; analyzed the tasks and performances of, and gave feedback to; other contributors in the project.
- Worked on over 10+ projects as an AI Trainer and Reviewer, including code refinement, evaluation tasks, and peer review at Outlier.ai.

Fellowship.AI

September 2025 – November 2025

Machine Learning Fellow

- Selected as a Fellow in Cohort 35 of the competitive Fellowship program by Launchpad.ai.
- Worked on the Real Estate Listings Optimization track, designing and implementing a Supabase database to manage realtor, property, location, and analysis metadata.
- Developed a robust data pipeline for parsing, cleaning, and uploading scraped listings at scale.
- Experimented with multiple scraping strategies; raw HTML parsing, Firecrawl-based extraction, and image-based techniques to construct a unified, structured database of real estate listings and realtor profiles.

GenMorphics AI Solution

January 2024 – October 2024

Quality Assurance Analyst

- Worked as a Quality Assurance Analyst for fine-tuning an LLM on SuperAnnotate's Project X and OctopusMath.
- Performed detailed annotation and quality assurance tasks across domains including data structures and algorithms, C++, Python, physics, and chemistry.
- Identified systematic issues in model outputs and contributed to creating high-quality, domain-balanced datasets for model training and evaluation.
- Ensured accuracy, clarity, and conceptual correctness of AI-generated responses for educational and STEM-focused applications.

Udvash-Ummesh

August 2022 – July 2024

Academic Materials Contributor

- Contributed to creating question sets for University Admission Mock Tests for the largest and most selective admission coaching organization in the country.
- Helped with the writing and translation of educational materials and practice problems for STEM subjects for Secondary, High Secondary and Admission Levels.
- Tutored students preparing for university entrance exams, helping them strengthen problem-solving skills and understand complex concepts.
- Contributed to one-on-one mentoring, providing guidance on exam strategies and effective study techniques.

Honors and Achievements

Champion

September 2025

Bangladesh University of Professionals CTF
Inter-University Cybersecurity Event (Capture-The-Flag)

Champion

May 2025

University of Asia Pacific CyberSiege
Inter-University Cybersecurity Event (Capture-The-Flag)

Champion

November 2024

East West University NRFCTF
Inter-University Cybersecurity Event (Capture-The-Flag)

Champion

October 2024

Cyber Crime Awareness Foundation CyberHackathon
National Cybersecurity Event (Capture-The-Flag)

Champion

July 2022

Bangladesh University of Professionals Film Festival
Competitive Script Writing Event

Runner's Up

May 2025

University of Asia Pacific TechTron
Poster Presentation (Single Image Generative Adversarial Networks)

2nd Runner's Up

November 2023

University of Dhaka CTF
Intra-University Cybersecurity Event (Capture-The-Flag)

3rd Runner's Up

November 2024

Three Minute Thesis - Presentation (TMT-P)
Research Presentation (Natural Language Processing)

3rd Runner's Up

October 2024

Bangladesh University of Engineering and Technology CTF
National Cybersecurity Event (Capture-The-Flag)

3rd Runner's Up

April 2024

Islamic University of Technology CTF
Inter-University Cybersecurity Event (Capture-The-Flag)

4th Runner's Up

February 2025

Bangladesh Armed Forces CyberRaid
National Cybersecurity Event (Capture-The-Flag); team was later interviewed as a part of the recruitment process of the Bangladesh Armed Forces IT Directorate

7th Place

April 2024

Islamic University of Technology Code Refactoring Contest
Competitive Programming Event (Java)

Finalist

March 2023

First Dhaka University Research Society Undergraduate Research Summit
Selected among top 26 research projects from 200+ submissions for undergraduate research presentation

Earlier Achievements

2018-2022

National Competitions in STEM & Humanities

Earned 30+ awards across writing, mathematics, physics, and academic quizzing including, but not limited to, representing institution on national television quiz programs. Notable awards include:

- **Champion** - National Cultural Classicists Carnival 2020
- **Silver Honor** - International Youth Math Challenge (IYMC) 2019
- **Champion** - 1st Yugen Scientific Write-Up Competition 2019
- **Champion** - Adamjee Mathematician Hunt 2018
- **2nd Runner's up** - Bangladesh Television National Quiz Competition (BTV Quiz Quiz) 2018

Technical Skills

Languages: C/C++, Python, Java, Swift, Rust, ARM Assembly, Dart, SQL, JavaScript, TypeScript, Bash;

Frontend: HTML/CSS, Vue, React, NextJS, JavaFX, Flutter;

Backend: NodeJS, Django, FastAPI, RESTful APIs;

Databases: PostgreSQL, Supabase, Firebase;

Machine Learning & AI: PyTorch, TensorFlow, Keras, scikit-learn, spaCy, Hugging Face;

LLM & Generative AI: RAG Architecture, LLM Finetuning, Prompt Engineering, OpenAI & Gemini APIs, LangChain

Agentic AI & Automation: Playwright, LangManus;

Networking: TCP/UDP, distributed systems, DHT, chunked streaming, Network Simulation (CISCO);

DevOps & Tools: Qt, Git, Docker, Railway, LaTeX, Firebase, AWS, Vercel;

Security & Forensics: Encryption/Cryptography, Penetration Testing (Burp Suite), Digital Forensics (Wireshark, FTK).

External Courses

Introduction to Machine Learning

2021

Duke University (Coursera)

Qt 5 C++ GUI Development for Beginners: The Fundamentals

2023

Udemy